

Large series priors for adaptive and optimal Bayesian nonparametric regression

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Abstract

Random series provide a powerful framework for nonparametric Bayesian inference and are widely used. A key parameter in their application is the number of basis functions (i.e., truncation level or series order), which critically influences performance. However, selecting this parameter is challenging: fixed choices are computationally simple but may lack robustness in finite samples and theoretical justification asymptotically, while hierarchical approaches using hyperpriors offer theoretical guarantees but often suffer from slow posterior computation, which limits practical use. In this article, we propose a novel approach using a large but deterministic number of basis functions, scaling as n^b for a constant $b > 0$ (where n is the sample size), alongside shrinkage priors on the coefficients. This large random series prior combines rich representational capacity with automated regularization embedded in adaptive shrinkage priors. On the theoretical front, we establish posterior contraction rates for general shrinkage priors and basis systems, and show that a single deterministic product prior achieves pointwise adaptation and near-minimax rates over the smoothness range covered by dictionary growth and basis order fixed in advance. Computationally, the large prior avoids the complexities of varying-dimensional parameter spaces and enables efficient posterior sampling using standard algorithms for shrinkage priors. Numerical experiments illustrate its flexibility and accuracy.

Keywords: Bayesian nonparametrics, random series prior, adaptivity, shrinkage priors, sparsity, posterior contraction, minimax optimality

1 Introduction

Random series priors are among the most popular choices to construct nonparametric priors in Bayesian inference. Consider the regression model given by

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad (1)$$

for $i = 1, \dots, n$. With a given series dictionary $\{\xi_j\}_{j \geq 1}$, the random series prior induces random functions $f(x) = \sum_{j=1}^{J_n} \theta_j \xi_j(x)$ with $\theta = (\theta_1, \dots, \theta_{J_n})$ being the regression coefficients drawn from a prior distribution $\theta \in \Pi(\cdot)$. Here J_n , the number of basis functions, balances the bias-variance trade-off and is critical in driving the performance of random series, a phenomenon frequently observed in empirical studies and relatively well understood theoretically. Several strategies for selecting J_n have been explored in the literature, as described below, which face significant challenges.

Methods based on cross-validation, information criteria, or empirical Bayes [Leung \(2005\)](#); [Ruppert \(2002\)](#); [Liu and Dobriban \(2019\)](#) tend to suffer from computational demands, limited flexibility, and sensitivity to outliers. Additionally, rigorous theoretical understanding of these selection procedures remains limited, particularly in terms of their adaptivity to a wide range of functions with unknown smoothness. As such, heuristically observed superior performance in a particular application might not generalize well to other settings, and the same strategy, when transferred to a different scenario, can potentially result in substantial accuracy loss.

Hierarchical Bayesian methods address some of these issues by placing a prior distribution directly on J_n , offering a principled way to quantify uncertainty in model complexity. This fully Bayesian treatment enjoys strong theoretical guarantees, with [Shen and Ghosal \(2015\)](#) establishing posterior adaptivity and optimality. In a different context for finite mixture models, [Guha et al. \(2021\)](#) established adaptive posterior contraction rates. However, hierarchical approaches face computational challenges due to dimensional variability.

Meanwhile, the Bayesian shrinkage literature has established strong contraction properties without putting a prior on model dimension. [van der Pas et al. \(2017\)](#) obtained adaptive, near-minimax contraction for the Horseshoe in the sparse normal-means model, and [Wei and Ghosal \(2020\)](#) derived contraction results for shrinkage priors in high-dimensional logistic regression. Related work on Bayesian penalized splines establishes posterior contraction under smoothing priors ([Bach and Klein, 2021](#)), while [Bühlmann and van de Geer \(2011\)](#) provide a broader treatment of sparsity and regularization in high-dimensional models. Although these results arise in different settings, they motivate the use of a large fixed dictionary with adaptive shrinkage in the present nonparametric problem.

In this article, we aim to borrow the best of both worlds: the computational simplicity of fixed- J_n methods and the theoretical adaptivity typically associated with hierarchical approaches. Our method attains adaptivity by using a non-stochastic choice of J_n while preserving the simplicity of fixed-series formulations. We set $J_n = n^b$ for some $b > 0$ and place shrinkage priors on the resulting high-dimensional parameter spaces. This large series prior offers two main contributions to the nonparametric Bayesian literature: we establish posterior contraction rates for general shrinkage priors with large series representations, and we show that one prior, with dictionary growth fixed independently of the truth, is pointwise adaptive and near minimax optimal throughout its admissible smoothness range under the stated sparse-approximation and prior conditions. The numerical illustrations examine the finite-sample behavior of practical large-series implementations and are interpreted separately from these posterior-contraction guarantees.

Notation. Let c_n and d_n be positive sequences. We write $c_n \lesssim d_n$ if $c_n \leq C d_n$ eventually for a constant $C > 0$, $c_n \asymp d_n$ if both $c_n \lesssim d_n$ and $d_n \lesssim c_n$, and $c_n = o(d_n)$ if $c_n/d_n \rightarrow 0$. For a vector $\theta \in \mathbb{R}^{J_n}$, write $\|\theta\|_p = (\sum_j |\theta_j|^p)^{1/p}$. For a function g , let

$$\|g\|_{p,n} = \left\{ \frac{1}{n} \sum_{i=1}^n |g(x_i)|^p \right\}^{1/p}, \quad d_n(g, h) = \|g - h\|_{2,n}.$$

We identify a function with its evaluation vector $(g(x_1), \dots, g(x_n)) \in \mathbb{R}^n$. All balls, projections, and coverings used below therefore take place in $\mathbb{R}^n$ equipped with the inner product $\langle u, v \rangle_n = n^{-1} \sum_i u_i v_i$, whose associated norm is $\|\cdot\|_{2,n}$.

Throughout, the design points $x_1, \dots, x_n$ are deterministic and the noise variance σ^2 is known. Write $Y^{(n)} = (y_1, \dots, y_n)$, and let P_0 and E_0 denote probability and expectation under Model (1) with regression function f_0 . For an evaluation vector $v \in \mathbb{R}^n$, let P_v and E_v denote probability and expectation when the observations are independent $N(v_i, \sigma^2)$. If v is the evaluation vector of a function f , we also write P_f and E_f . The symbols p_θ and p_0 denote the corresponding joint densities under f_θ and f_0 , respectively.

2 Large series prior and theoretical guarantees

For a specified basis system, such as B-splines or Fourier functions, we use a deterministic but large dimension $J_n \rightarrow \infty$; in the principal examples $J_n = \lceil n^b \rceil$ for a constant $b > 0$. All subsequent definitions and asymptotic statements are understood for n sufficiently large that $J_n \geq 2$. For $\theta = (\theta_1, \dots, \theta_{J_n}) \in \mathbb{R}^{J_n}$, set

$$f_\theta(x) = \sum_{j=1}^{J_n} \theta_j \xi_j(x) = \theta^T \xi(x).$$

For every n , the coefficient prior is the triangular-array product

$$\Pi_n = \bigotimes_{j=1}^{J_n} \Pi_{n,j}, \quad d\Pi_{n,j}(t) = \pi_n(t) dt,$$

where the common marginal density π_n may depend on n . We use Π_n also for the induced prior and posterior on $f = f_\theta$.

Let $\tilde{\epsilon}_n \downarrow 0$ be the prior concentration rate, let $a_n \in \{1, \dots, J_n\}$ be an effective sparsity level, let $\delta_n > 0$ be the negligibility threshold, and let $E_n > 0$ be a nondecreasing signal bound. Define

$$s_n = \max \left\{ a_n, \left\lceil \frac{n\tilde{\epsilon}_n^2}{\log J_n} \right\rceil \right\}, \quad \epsilon_n = \sqrt{\frac{s_n \log J_n}{n}}, \quad n\epsilon_n^2 = s_n \log J_n. \quad (2)$$

The standing sequence requirements are

$$n\tilde{\epsilon}_n^2 \rightarrow \infty, \quad \delta_n \leq \frac{\tilde{\epsilon}_n}{J_n}, \quad s_n \log J_n = o(n). \quad (3)$$

Consequently, $\epsilon_n \geq \tilde{\epsilon}_n$, $n\epsilon_n^2 \rightarrow \infty$, and $\epsilon_n \rightarrow 0$, while

$$\epsilon_n^2 \asymp \max \left\{ \tilde{\epsilon}_n^2, \frac{a_n \log J_n}{n} \right\}. \quad (4)$$

In particular, if $n\tilde{\epsilon}_n^2 \lesssim a_n \log J_n$, then $s_n \asymp a_n$ and $\epsilon_n \asymp \sqrt{a_n \log J_n / n}$.

Assumption 1 (Basis). There is a constant $K \geq 1$ such that

$$\max_{1 \leq j \leq J_n} \|\xi_j\|_{2,n} \leq K \quad (5)$$

for every n .

Assumption 2 (Sparse approximability). For all sufficiently large n , there exists $\theta_0 = \theta_0(n) \in \mathbb{R}^{J_n}$ such that

$$\|f_0 - f_{\theta_0}\|_{2,n} \leq \tilde{\epsilon}_n, \quad (6)$$

$$\#\{j : |\theta_{0,j}| > \delta_n\} \leq a_n, \quad \max_{1 \leq j \leq J_n} |\theta_{0,j}| \leq E_n. \quad (7)$$

For one such approximant, fix a set $\xi_n^* \subset \{1, \dots, J_n\}$ of exactly a_n indices containing every j for which $|\theta_{0,j}| > \delta_n$. Thus $|\theta_{0,j}| \leq \delta_n$ for every $j \notin \xi_n^*$.

Assumption 3 (Prior conditions of Wei–Ghosal). The coordinates are independent and identically distributed under Π_n with common density π_n . Assumption 2 is understood

to fix one admissible pair (θ_0, ξ_n^*) . Because the coordinate prior is iid and $|\xi_n^*| = a_n$, the probability in (WG4) depends only on $J_n - a_n$, not on the particular admissible set or on the values of θ_0 . Fix a constant $\eta \in (0, 1)$. The following adaptations of conditions (3)–(5) of Wei and Ghosal (2020) hold for all sufficiently large n :

$$\int_{|t| > \delta_n} \pi_n(t) dt \leq J_n^{-(1+\mu)} \quad \text{for a constant } \mu > 0, \quad (\text{WG3})$$

$$-\log \Pi_n \left(\sum_{j \notin \xi_n^*} |\theta_j| \leq \eta \tilde{\epsilon}_n \right) \leq c_4 n \tilde{\epsilon}_n^2 \quad \text{for a constant } c_4 \geq 0. \quad (\text{WG4})$$

Let

$$\underline{\pi}_{n, E_n} = \text{ess inf}_{|t| \leq E_n} \pi_n(t) > 0.$$

For a constant $c_5 \geq 0$, assume

$$a_n \left[\log a_n + 2 \log \left(\frac{1}{\tilde{\epsilon}_n} \right) - \log \underline{\pi}_{n, E_n} \right]_+ \leq c_5 n \tilde{\epsilon}_n^2. \quad (\text{WG5})$$

Condition (WG3) concentrates each marginal near zero and drives the sieve-complement bound. Condition (WG4) controls the joint contribution of the inactive coordinates to the local prior mass, whereas (WG5) ensures enough mass around the active coordinates. These are the nonparametric-series versions of conditions (3)–(5) of Wei and Ghosal (2020) under the substitutions $p \mapsto J_n$, $\bar{\epsilon}_n \mapsto \tilde{\epsilon}_n$, their threshold $a_n^{\text{WG}} \mapsto \delta_n$, $s_* \mapsto |\xi_n^*| = a_n$, and $E \mapsto E_n$. Their separate coefficient cutoff in condition (6), used for a coefficient-box entropy argument, is unnecessary here because the proof covers prediction-distance shells inside their low-dimensional evaluation subspaces.

Theorem 1. Suppose Model (1), Assumptions 1–3, and the standing requirements (3) hold. Then there exist a constant $M > 0$ and a fixed integer $A \geq 1$, depending only on $(K, \sigma, \mu, \eta, c_4, c_5)$, such that, for all sufficiently large n ,

$$E_0 \Pi_n \left(\|f_0 - f\|_{2,n} > M \epsilon_n \mid Y^{(n)} \right) \leq 6e^{-n \tilde{\epsilon}_n^2}. \quad (8)$$

Consequently,

$$\Pi_n \left(\|f_0 - f\|_{2,n} > M \epsilon_n \mid Y^{(n)} \right) \xrightarrow{P_0} 0. \quad (9)$$

Moreover,

$$E_0 \Pi_n \left(\#\{j : |\theta_j| > \delta_n\} > A s_n \mid Y^{(n)} \right) \leq 2e^{-n \tilde{\epsilon}_n^2}, \quad (10)$$

and the corresponding posterior probability converges to zero in P_0 -probability. The contraction bound and conclusion remain valid with any sequence $M_n \rightarrow \infty$ in place of M .

Similar theoretical guarantees exist in related settings. [Wei and Ghosal \(2020\)](#) established analogous contraction properties for shrinkage priors in parametric logistic regression, while [Song and Liang \(2023\)](#) analyzed high-dimensional linear regression with a number of covariates that may substantially exceed the sample size. Our setting is instead nonparametric: the sparse vector is an approximating series representation of f_0 , and Assumption 2 makes this structural requirement explicit.

The deterministic-scale product Horseshoe of [Carvalho et al. \(2009\)](#) satisfies (WG3)–(WG5) under the scale and signal-size conditions stated in the Appendix, as follows from the estimates in [Wei and Ghosal \(2020, Theorem 3.3\)](#). This assertion concerns a deterministic n -dependent global scale τ_n . A hierarchy with a shared random global scale is not covered by Theorem 1, because integrating out that scale makes the coordinates dependent.

The corollaries require a nested sparse approximation property in addition to the usual approximation error bound. The next two lemmas verify this property for the concrete dictionaries used below. They also make clear that the spline construction is a redundant union of resolution blocks, not a single oversized fine-grid basis.

For $\alpha > 0$, let $r(\alpha)$ be any integer strictly larger than α . Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and define

$$\Delta_{he_k}^r f(x) = \sum_{v=0}^r (-1)^{r-v} \binom{r}{v} f(x + vhe_k),$$

where addition is modulo one. For $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d)$, put

$$\|f\|_{\mathcal{H}_{\text{per}}^{\boldsymbol{\alpha}}} = \|f\|_{\infty} + \max_{1 \leq k \leq d} \sup_{0 < |h| \leq 1/2} |h|^{-\alpha_k} \|\Delta_{he_k}^{r(\alpha_k)} f\|_{\infty}.$$

This is the coordinatewise periodic anisotropic Hölder–Zygmund norm. For noninteger smoothness it is equivalent to the usual periodic Hölder norm, and the usual periodic Lipschitz–Hölder ball at an integer index is contained in the corresponding Hölder–Zygmund ball. Throughout, at an integer index $m \geq 1$, the notation C^m uses the Lipschitz–Hölder convention $C^m = C^{m-1,1}$.

Lemma 1 (Sparse approximation by the real Fourier dictionary). Let d be fixed, let every $\alpha_k > 0$, and suppose that the real-valued function f_0 satisfies

$$\|f_0\|_{\mathcal{H}_{\text{per}}^{\boldsymbol{\alpha}}} \leq R$$

for a fixed $R < \infty$. Define

$$\psi_0(x) = 1, \quad \psi_{2j-1}(x) = \sqrt{2} \cos(2\pi jx), \quad \psi_{2j}(x) = \sqrt{2} \sin(2\pi jx), \quad j \geq 1.$$

For integers $M_1, \dots, M_d \geq 1$, consider the tensor dictionary

$$\mathcal{F}_M = \left\{ \Psi_\nu(x) = \prod_{k=1}^d \psi_{\nu_k}(x_k) : 0 \leq \nu_k \leq 2M_k, \ 1 \leq k \leq d \right\}.$$

Its coordinate cardinalities are $L_k = 2M_k + 1$ and its total cardinality is $J_M = \prod_{k=1}^d L_k$. For every integer rectangle $\mathbf{m} = (m_1, \dots, m_d)$ with $1 \leq m_k \leq L_k$, there is a coefficient vector $\theta(\mathbf{m}) \in \mathbb{R}^{J_M}$ such that

$$\begin{aligned} \#\text{supp } \theta(\mathbf{m}) &\leq \prod_{k=1}^d m_k, \\ \|f_0 - f_{\theta(\mathbf{m})}\|_\infty &\leq CR \sum_{k=1}^d m_k^{-\alpha_k}, \\ \|\theta(\mathbf{m})\|_\infty &\leq R. \end{aligned} \tag{11}$$

Here C depends only on $(d, \boldsymbol{\alpha})$. Consequently the same error bound holds in $\|\cdot\|_{2,n}$, and

$$\max_{\Psi \in \mathcal{F}_M} \|\Psi\|_{2,n} \leq 2^{d/2}.$$

For $d = 1$ this is the univariate sparse profile used below; taking $\alpha_k = \alpha$ and $m_k = m$ gives the isotropic profile; and the full rectangular conclusion gives the anisotropic profile.

Proof. Let σ_N be the N th Fejér mean and put

$$V_q = 2\sigma_{2q-1} - \sigma_{q-1}, \quad q \geq 1.$$

The operator V_q reproduces trigonometric polynomials of degree at most q , has output degree at most $2q - 1$, has Fourier multipliers in $[0, 1]$, and satisfies $\|V_q\|_{\infty \rightarrow \infty} \leq 3$. For an integer atom budget m , let $Q_m = \sigma_0$ when $m \leq 2$, and for $m \geq 3$ put

$$q_m = \left\lfloor \frac{m+1}{4} \right\rfloor, \quad Q_m = V_{q_m}.$$

For $m \geq 3$, one has $q_m \geq m/8$. Moreover, $Q_m g$ uses at most $4q_m - 1 \leq m$ real Fourier atoms. The periodic Jackson inequality [Temlyakov \(2018\)](#), followed by the reproduction property of V_{q_m} , gives

$$\|g - Q_m g\|_\infty \leq C_\alpha R m^{-\alpha} \tag{12}$$

whenever the corresponding one-dimensional Hölder–Zygmund norm of g is at most R . The cases $m = 1, 2$ are absorbed by increasing the constant.

Let $Q_{m_k}^{(k)}$ act in the k th coordinate and set

$$P_m f_0 = Q_{m_1}^{(1)} \cdots Q_{m_d}^{(d)} f_0.$$

The operators commute, and the telescoping identity

$$I - \prod_{k=1}^d Q_{m_k}^{(k)} = \sum_{k=1}^d \left(\prod_{v < k} Q_{m_v}^{(v)} \right) (I - Q_{m_k}^{(k)})$$

together with (12) yields

$$\|f_0 - P_m f_0\|_\infty \leq CR \sum_{k=1}^d m_k^{-\alpha_k}.$$

The output uses at most m_k atoms in coordinate k , hence at most $\prod_k m_k$ tensor atoms. Since $m_k \leq 2M_k + 1$, its largest frequency does not exceed M_k , so these atoms belong to $\mathcal{F}_M$.

Every nonzero real Fourier coefficient has the form

$$\left\{ \prod_{k=1}^d \lambda_{m_k, \nu_k} \right\} \int_{\mathbb{T}^d} f_0(x) \Psi_\nu(x) dx, \quad 0 \leq \lambda_{m_k, \nu_k} \leq 1.$$

Because $\|\psi_0\|_1 = 1$ and $\|\psi_j\|_1 = 2\sqrt{2}/\pi < 1$ for $j \geq 1$, its absolute value is at most $\|f_0\|_\infty \leq R$. Finally, $\|g\|_{2,n} \leq \|g\|_\infty$ and $\|\Psi_\nu\|_\infty \leq 2^{d/2}$, proving the remaining claims. $\square$

Lemma 2 (Sparse approximation by multiresolution B-splines). Fix an integer spline order $q \geq 1$, independent of n , where an order- q spline has polynomial degree at most $q - 1$. At level ℓ , divide $[0, 1]$ into $K_\ell = 2^\ell$ equal subintervals and let

$$\mathcal{B}_\ell^{(q)} = \{B_{\ell,1}, \dots, B_{\ell,D_\ell}\}, \quad D_\ell = K_\ell + q - 1,$$

be the standard open-knot, partition-of-unity B-spline basis. Define the tagged multiresolution union

$$\mathcal{U}_H^{(q)} = \bigsqcup_{\ell=0}^H \mathcal{B}_\ell^{(q)}, \quad U_H = \sum_{\ell=0}^H D_\ell.$$

Atoms from different resolutions retain different coefficient indices. For fixed d and levels

$H_1, \dots, H_d$, form

$$\mathcal{D}_{\mathbf{H}}^{(q)} = \bigotimes_{k=1}^d \mathcal{U}_{H_k}^{(q)}.$$

Its coordinate cardinalities are $L_k = U_{H_k}$ and its total cardinality is $J_{\mathbf{H}} = \prod_{k=1}^d L_k$. Every atom takes values in $[0, 1]$, so its empirical norm is at most one.

The following conclusions hold with constants independent of $\mathbf{H}$, the selected integer truncation levels, n , and the design points:

1. If $d = 1$, $0 < \alpha \leq q$, and f_0 belongs to a fixed ball in $C^\alpha([0, 1])$, then for every $1 \leq m \leq U_H$ there is $\theta(m) \in \mathbb{R}^{U_H}$ satisfying

$$\#\text{supp } \theta(m) \leq C_{\text{sp}} m, \quad \|f_0 - f_{\theta(m)}\|_\infty \leq C_{\text{app}} m^{-\alpha}, \quad \|\theta(m)\|_\infty \leq E.$$

2. If d is fixed, $0 < \alpha \leq q$, and f_0 belongs to a fixed ball in $C^\alpha([0, 1]^d)$, then, when $H_1 = \dots = H_d = H$, for every $1 \leq m \leq U_H$ there is $\theta(m) \in \mathbb{R}^{U_H^d}$ satisfying

$$\#\text{supp } \theta(m) \leq C_{\text{sp}} m^d, \quad \|f_0 - f_{\theta(m)}\|_\infty \leq C_{\text{app}} m^{-\alpha}, \quad \|\theta(m)\|_\infty \leq E.$$

3. If $\alpha_1, \dots, \alpha_d$ are positive integers with $\max_k \alpha_k \leq q$ and f_0 belongs to a fixed ball in the anisotropic Hölder class of (Shen and Ghosal, 2015, Section 3.2 and Lemma 2), then for every rectangle $1 \leq m_k \leq U_{H_k}$ there is $\theta(\mathbf{m}) \in \mathbb{R}^{J_{\mathbf{H}}}$ satisfying

$$\begin{aligned} \#\text{supp } \theta(\mathbf{m}) &\leq C_{\text{sp}} \prod_{k=1}^d m_k, \\ \|f_0 - f_{\theta(\mathbf{m})}\|_\infty &\leq C_{\text{app}} \sum_{k=1}^d m_k^{-\alpha_k}, \quad \|\theta(\mathbf{m})\|_\infty \leq E. \end{aligned}$$

In every case the same approximation bound holds in empirical norm. Thus the three conclusions give, respectively, the univariate, isotropic, and, in the stated integer-smoothness regime, anisotropic sparse profiles used below.

Proof. Because q is fixed,

$$D_H \leq U_H = \sum_{\ell=0}^H D_\ell \leq C_q D_H, \quad U_H \asymp 2^H. \quad (13)$$

For every $1 \leq m \leq U_H$, choose a level $\ell_H(m) \leq H$ whose dimension is nearest to m from the geometric sequence $D_0, \dots, D_H$, using level zero below D_0 and level H above D_H . Since

$D_{\ell+1} \leq 2D_\ell$, fixed constants $0 < c_q < C'_q < \infty$ satisfy

$$c_q m \leq D_{\ell_H(m)} \leq C'_q m, \quad 1 \leq m \leq U_H. \quad (14)$$

This establishes the required comparison for every integer m , not only for dyadic values.

At a fixed resolution, the spline approximation results in (Shen and Ghosal, 2015, Lemmas 1–2 and Remark 7) give the univariate and isotropic sup-norm errors displayed above. Their dual-basis stability argument also permits the coefficients to be chosen uniformly bounded by a constant depending only on the fixed Hölder ball, q , and d . Select the block at level $\ell_H(m)$ and set all coefficients in the other resolution blocks equal to zero. Equations (13)–(14) then give support $O(m)$ in one dimension and $O(m^d)$ in the isotropic tensor case, while preserving error $O(m^{-\alpha})$ and the coefficient bound.

For the anisotropic conclusion, apply (Shen and Ghosal, 2015, Lemma 2) at the rectangular block with coordinate levels $\ell_{H_k}(m_k)$. Its sup-norm error is bounded by a constant multiple of $\sum_k D_{\ell_{H_k}(m_k)}^{-\alpha_k}$, its coefficient vector is uniformly bounded by the tensor dual-basis argument in that proof, and its support is $\prod_k D_{\ell_{H_k}(m_k)} = O(\prod_k m_k)$. Padding every coefficient outside the selected block with zero proves the claim. Finally, $0 \leq B_{\ell,j} \leq 1$ and $\|g\|_{2,n} \leq \|g\|_\infty$ give the basis and empirical-norm bounds. $\square$

The integer restriction in the anisotropic spline conclusion is precisely the restriction in (Shen and Ghosal, 2015, Lemma 2); their Remark 7 removes it only in the isotropic case. Accordingly, Corollary 3 continues to allow arbitrary positive real α_k through its explicit sparse-profile hypothesis and through Lemma 1, but Lemma 2 supplies its B-spline specialization only in the stated regime. Extending that specialization to fractional anisotropic smoothness requires a separate stable anisotropic spline quasi-interpolation result.

For the asymptotic dictionaries in the corollaries, choose the Fourier cutoffs or the finest spline levels so that the coordinate cardinalities have the prescribed polynomial orders. This is possible because $2M_k + 1 \asymp M_k$ for Fourier blocks and $U_{H_k} \asymp 2^{H_k}$ for the multiresolution spline unions.

Corollary 1. Let f_0 belong to a fixed ball in $C^\alpha([0, 1])$. When a Fourier dictionary is used, assume that f_0 has a periodic extension belonging to a fixed ball in $\mathcal{H}_{\text{per}}^\alpha(\mathbb{T})$. Suppose $J_n \asymp n^b$ with

$$b \geq \frac{1}{2\alpha + 1},$$

Assumption 1 holds, and, for every integer $m \leq J_n$, there is a vector $\theta(m) \in \mathbb{R}^{J_n}$ such that

$$\# \text{supp } \theta(m) \leq C_{\text{sp}} m, \quad \|f_0 - f_{\theta(m)}\|_{2,n} \leq C_{\text{app}} m^{-\alpha}, \quad \|\theta(m)\|_{\infty} \leq E. \quad (15)$$

where $C_{\text{sp}}, C_{\text{app}}, E$ are fixed positive constants independent of n and m . Put

$$N_n = \frac{n}{\log J_n}, \quad m_n = \left\lceil N_n^{1/(2\alpha+1)} \right\rceil, \quad a_n = \lceil C_{\text{sp}} m_n \rceil, \quad (16)$$

$$\tilde{\epsilon}_n = C_{\epsilon} N_n^{-\alpha/(2\alpha+1)}, \quad E_n = E, \quad \delta_n = c_{\delta} \frac{\tilde{\epsilon}_n}{J_n}, \quad (17)$$

where C_{ϵ} is sufficiently large and $c_{\delta} \in (0, 1)$. Suppose the coefficient prior satisfies Assumption 3. In particular, the deterministic-scale product Horseshoe described in the Appendix, with scale (30), is admissible.

Then $s_n \asymp a_n$ and

$$a_n \asymp \left(\frac{n}{\log n} \right)^{1/(2\alpha+1)}, \quad \epsilon_n \asymp \left(\frac{n}{\log n} \right)^{-\alpha/(2\alpha+1)}.$$

Consequently, for constants $M, A', c > 0$,

$$E_0 \Pi_n \left(\|f - f_0\|_{2,n} > M \left(\frac{n}{\log n} \right)^{-\alpha/(2\alpha+1)} \middle| Y^{(n)} \right) \leq 6e^{-ca_n \log J_n}, \quad (18)$$

$$E_0 \Pi_n \left(\#\{j : |\theta_j| > \delta_n\} > A' a_n \middle| Y^{(n)} \right) \leq 2e^{-ca_n \log J_n}, \quad (19)$$

and both posterior probabilities converge to zero in P_0 -probability.

Corollary 2. Let d be fixed and let f_0 belong to a fixed ball in $C^{\alpha}([0, 1]^d)$. For a tensor Fourier dictionary, assume that f_0 has a coordinatewise periodic extension belonging to a fixed ball in $\mathcal{H}_{\text{per}}^{(\alpha, \dots, \alpha)}(\mathbb{T}^d)$. Let the large dictionary have coordinate size $L_n \asymp n^b$ and total size $J_n \asymp L_n^d$, where

$$b \geq \frac{1}{2\alpha + d}.$$

Suppose Assumption 1 holds and, for every integer $m \leq L_n$, there is a vector $\theta(m) \in \mathbb{R}^{J_n}$ such that

$$\# \text{supp } \theta(m) \leq C_{\text{sp}} m^d, \quad \|f_0 - f_{\theta(m)}\|_{2,n} \leq C_{\text{app}} m^{-\alpha}, \quad \|\theta(m)\|_{\infty} \leq E. \quad (20)$$

where $C_{\text{sp}}, C_{\text{app}}, E$ are fixed positive constants independent of n and m . Define

$$N_n = \frac{n}{\log J_n}, \quad m_n = \left\lceil N_n^{1/(2\alpha+d)} \right\rceil, \quad a_n = \lceil C_{\text{sp}} m_n^d \rceil, \quad (21)$$

$$\tilde{\epsilon}_n = C_\epsilon N_n^{-\alpha/(2\alpha+d)}, \quad E_n = E, \quad \delta_n = c_\delta \frac{\tilde{\epsilon}_n}{J_n}. \quad (22)$$

Here C_ϵ is sufficiently large and $c_\delta \in (0, 1)$. Suppose the coefficient prior satisfies Assumption 3; the deterministic-scale product Horseshoe in the Appendix is one such prior.

Then $s_n \asymp a_n$ and

$$a_n \asymp \left(\frac{n}{\log n} \right)^{d/(2\alpha+d)}, \quad \epsilon_n \asymp \left(\frac{n}{\log n} \right)^{-\alpha/(2\alpha+d)}.$$

For constants $M, A', c > 0$, the expectation and P_0 -probability conclusions of Theorem 1 hold with this contraction rate and with effective-support bound $A'a_n$; the two expectation bounds are at most $6e^{-ca_n \log J_n}$ and $2e^{-ca_n \log J_n}$, respectively.

Corollary 3. Let d be fixed, let $R > 0$, and let $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d) \in (0, \infty)^d$. For $x_{-k} \in [0, 1]^{d-1}$, write $f_{0,k,x_{-k}}(t) = f_0(x_1, \dots, x_{k-1}, t, x_{k+1}, \dots, x_d)$. Assume that f_0 belongs to the fixed anisotropic Hölder ball $C^\alpha([0, 1]^d; R)$ defined by

$$\max_{1 \leq k \leq d} \sup_{x_{-k} \in [0, 1]^{d-1}} \|f_{0,k,x_{-k}}\|_{C^{\alpha_k}([0, 1])} \leq R. \quad (23)$$

For a tensor Fourier dictionary, assume additionally that f_0 has a coordinatewise periodic extension belonging to a fixed ball in $\mathcal{H}_{\text{per}}^\alpha(\mathbb{T}^d)$. Define

$$\alpha^* = \frac{d}{\sum_{k=1}^d \alpha_k^{-1}}, \quad r = \frac{\alpha^*}{2\alpha^* + d}, \quad D = \frac{d}{2\alpha^* + d}, \quad (24)$$

$$\beta_k = \frac{\alpha^*}{\alpha_k(2\alpha^* + d)}, \quad k = 1, \dots, d. \quad (25)$$

Let the full rectangular dictionary have coordinate sizes $L_{n,k} \asymp n^{b_k}$ for fixed constants $b_k \geq \beta_k$ and total size $J_n \asymp \prod_{k=1}^d L_{n,k}$. Suppose Assumption 1 holds and, for every integer rectangle $\mathbf{m} = (m_1, \dots, m_d)$ with $m_k \leq L_{n,k}$, there is a vector $\theta(\mathbf{m}) \in \mathbb{R}^{J_n}$ satisfying

$$\begin{aligned} \#\text{supp } \theta(\mathbf{m}) &\leq C_{\text{sp}} \prod_{k=1}^d m_k, \\ \|f_0 - f_{\theta(\mathbf{m})}\|_{2,n} &\leq C_{\text{app}} \sum_{k=1}^d m_k^{-\alpha_k}, \\ \|\theta(\mathbf{m})\|_\infty &\leq E. \end{aligned} \quad (26)$$

Here $C_{\text{sp}}, C_{\text{app}}, E$ are fixed positive constants independent of n and $\mathbf{m}$, although they may

depend on (d, R, α) and on fixed basis orders. Put

$$N_n = \frac{n}{\log J_n}, \quad m_{n,k} = \lceil N_n^{\beta_k} \rceil, \quad a_n = \left\lceil C_{\text{sp}} \prod_{k=1}^d m_{n,k} \right\rceil, \quad (27)$$

$$\tilde{\epsilon}_n = C_\epsilon N_n^{-r}, \quad E_n = E, \quad \delta_n = c_\delta \frac{\tilde{\epsilon}_n}{J_n}. \quad (28)$$

Here C_ϵ is sufficiently large and $c_\delta \in (0, 1)$. Suppose the coefficient prior satisfies Assumption 3; again, the Appendix supplies a concrete deterministic-scale product Horseshoe choice.

Then $s_n \asymp a_n$ and

$$a_n \asymp \left(\frac{n}{\log n} \right)^{d/(2\alpha^*+d)}, \quad \epsilon_n \asymp \left(\frac{n}{\log n} \right)^{-\alpha^*/(2\alpha^*+d)}.$$

For constants $M, A', c > 0$, the expectation and P_0 -probability conclusions of Theorem 1 hold with the displayed contraction rate and effective-support bound $A'a_n$; the two expectation bounds are at most $6e^{-ca_n \log J_n}$ and $2e^{-ca_n \log J_n}$, respectively.

Remark 1 (Scope of adaptation). Fix the dictionary growth exponents and the deterministic product-Horseshoe prior before observing the truth. In particular, for fixed $\kappa > \mu$ and $\rho > 0$, take

$$\tau_n = \frac{J_n^{-(2+\kappa+\rho)}}{\sqrt{n}}$$

as in (30). Once J_n is fixed, this prior law contains no unknown smoothness parameter. The smoothness-dependent sequences a_n , $\tilde{\epsilon}_n$, and δ_n in the corollaries are analytical sequences used to verify Theorem 1; they are not inputs to the prior or posterior.

In the univariate setting, one fixed exponent b therefore gives pointwise adaptation over

$$\mathcal{A}_b^{(1)} = \left\{ \alpha > 0 : b \geq \frac{1}{2\alpha + 1} \right\}.$$

For $0 < b < 1$ this is equivalently $\alpha \geq (1/b - 1)/2$, while $b \geq 1$ covers every fixed $\alpha > 0$. Choosing b separately as a function of the true α would instead be oracle tuning. In the isotropic d -dimensional setting, the analogous range is

$$\mathcal{A}_{b,d}^{\text{iso}} = \left\{ \alpha > 0 : b \geq \frac{1}{2\alpha + d} \right\}.$$

If $b < 1/d$, this means $\alpha \geq (1/b - d)/2$; if $b \geq 1/d$, every fixed $\alpha > 0$ is covered. For fixed

anisotropic exponents $\mathbf{b} = (b_1, \dots, b_d)$, the admissible range is

$$\mathcal{A}_{\mathbf{b}}^{\text{aniso}} = \left\{ \boldsymbol{\alpha} \in (0, \infty)^d : \frac{\alpha^*}{\alpha_k(2\alpha^* + d)} \leq b_k \text{ for every } k \right\}.$$

Thus a prescribed anisotropic set $\mathcal{A}$ is covered by choosing b_k at least the supremum of the displayed fraction over $\boldsymbol{\alpha} \in \mathcal{A}$; taking every $b_k \geq 1$ covers every fixed positive smoothness vector, although the resulting dictionary can be very large.

Periodic Fourier dictionaries impose no finite approximation-order restriction, subject to the periodic Hölder–Zygmund regularity stated above. For multiresolution B-splines, the order must also be fixed before the truth. Lemma 2 covers a prescribed univariate or isotropic range at most a fixed order q , while its anisotropic conclusion, under the cited Shen–Ghosal result, covers positive integer coordinate smoothness at most q . A finite-order spline dictionary therefore does not adapt over an unbounded smoothness range, and fractional anisotropic spline adaptation requires the additional approximation result described after that lemma.

These are pointwise adaptation statements: the same prior gives the stated rate for every fixed admissible smoothness index and fixed truth, but the constants may depend on that index and on the Hölder-ball radius. No bound uniform jointly over the smoothness index and the truth is asserted.

Remark 2 (Simpler polynomial choices). The preceding log-balanced choices give the rates obtained by balancing approximation and stochastic complexity in Theorem 1. More generally, if $J_n \asymp n^B$, $a_n \asymp n^\gamma$ with $\gamma < 1$, and $n\tilde{\epsilon}_n^2 \lesssim a_n \log J_n$, then $s_n \asymp a_n$ and Theorem 1 gives the polynomial-rate heuristic $\epsilon_n \asymp \sqrt{\log n} n^{(\gamma-1)/2}$, provided the displayed sparse-approximation, prior, and standing conditions hold. For comparison, choosing the simpler polynomial active dimensions gives

$$\begin{array}{lll} d = 1 : & a_n \asymp n^{1/(2\alpha+1)}, & \epsilon_n \asymp \sqrt{\log n} n^{-\alpha/(2\alpha+1)}, \\ \text{isotropic} : & a_n \asymp n^{d/(2\alpha+d)}, & \epsilon_n \asymp \sqrt{\log n} n^{-\alpha/(2\alpha+d)}, \\ \text{anisotropic} : & a_n \asymp n^{d/(2\alpha^*+d)}, & \epsilon_n \asymp \sqrt{\log n} n^{-\alpha^*/(2\alpha^*+d)}. \end{array}$$

The numerator d in the last two active-dimension exponents is essential: it is the dimension of the tensor approximation block.

Remark 3 (Why the B-spline formulation matters). For a single fine-grid B-spline basis, ordinary Hölder approximation theory does not by itself supply an a_n -sparse coefficient vector when $a_n \ll J_n$. Indeed, the canonical partition-of-unity representation of the constant function is $1 = \sum_{j=1}^{J_n} B_j$, with all coefficients equal to one, and hence is dense.

This observation does not rule out a different representation on a particular finite design. The corollaries use either a genuinely nested Fourier system, a multiresolution B-spline dictionary containing the selected coarse block, or any other dictionary satisfying the displayed sparse approximation profiles. Redundancy of a multiresolution dictionary is harmless for Theorem 1, whose proof uses only the uniform empirical norm bound and not linear independence.

3 Numerical Illustrations

This section examines the finite-sample behavior of large fixed dictionaries with Horseshoe shrinkage. The calculations are illustrations of the computational idea rather than numerical proofs of Theorem 1. In particular, the theorem concerns contraction of posterior mass under an iid product prior and a nested sparse-approximation condition. The computations use a practical hierarchical Horseshoe with a shared random global scale; after that scale is integrated out, the coefficients are dependent. The numerical B-spline basis is also an ordinary single-resolution basis, not the multiresolution dictionary used to justify the spline corollaries. Finally, the RMSE of a posterior point estimator is distinct from the posterior tail probability in (9). These distinctions guide the interpretation below.

3.1 Finite-sample comparison across benchmark functions

We consider three piecewise-polynomial regression functions on $[0, 1]$:

$$\begin{aligned} f_1(x) &= |x - 0.5|, \\ f_2(x) &= \frac{3}{2}|x - 0.1|^3 - 5|x - 0.4|^3, \\ f_3(x) &= \frac{1}{2}|x - 0.2|^5 + 2|x - 0.5|^5 - \frac{1}{2}|x - 0.9|^5. \end{aligned}$$

The functions have kinks in derivatives of orders 0, 2, and 4, respectively. Under the integer Lipschitz–Hölder convention $C^m = C^{m-1,1}$, their nominal regularities are therefore $m = 1, 3, 5$. This convention matters at integer smoothness indices. Moreover, $f_2(0) \neq f_2(1)$ and $f_3(0) \neq f_3(1)$, so these two functions do not satisfy the periodicity condition in the Fourier version of Corollary 1. The experiments compare finite-sample regularization across targets of increasing nominal regularity; they do not verify adaptation over the Hölder classes in that corollary.

For each benchmark, observations are generated on an equally spaced grid from the Gaussian regression model (1), with a noise level fixed within the experiment and scaled to the magnitude of the target. The initial comparison uses $n = 1000$ and the common dictionary-

size rule

$$J_n = \lfloor n^{0.7} \rfloor = 125.$$

The noise variance is treated as known. Posterior calculations are carried out in R using the `horseshoe` package and the hierarchical Horseshoe formulation of [Carvalho et al. \(2009\)](#). Because the global scale is shared and random, this computational prior is not the deterministic-scale product Horseshoe covered by Theorem 1.

As a dense global- ℓ_2 -shrinkage benchmark, we also fit ridge regression. If $X \in \mathbb{R}^{n \times p_n}$ denotes the relevant design matrix, where p_n is its actual number of columns, the ridge coefficient estimate has the form

$$\hat{\theta}_{\text{ridge}} = \arg \min_{\theta} \left\{ \|Y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 \right\},$$

with λ selected by 20-fold cross-validation; see [Friedman et al. \(2010\)](#); [Krstajic et al. \(2014\)](#); [Arlot and Celisse \(2010\)](#). Thus, within each Fourier fit, the comparison is between global-local Horseshoe shrinkage and dense global shrinkage under the same dictionary.

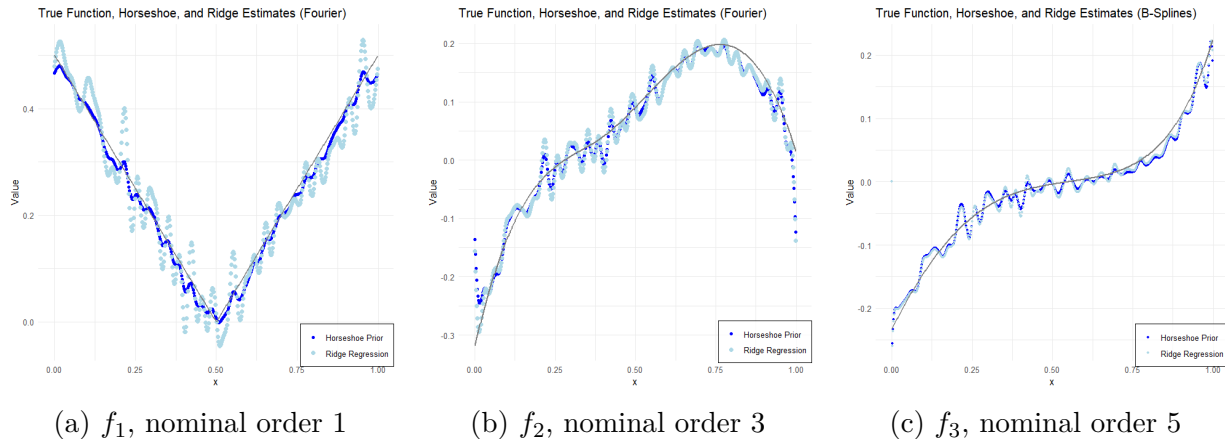


Figure 1: Fourier fits for the three benchmark functions, comparing the displayed Horseshoe point estimate with cross-validated ridge in one simulated data set. All panels use $n = 1000$ and $J_n = \lfloor n^{0.7} \rfloor = 125$.

Figure 1 gives a qualitative comparison for the displayed realizations. The Horseshoe curves exhibit less visible oscillation than ridge in several regions. Because the targets and their signal-to-noise scales differ across panels, this comparison cannot by itself attribute the differences to smoothness or establish adaptation.

We next hold $n = 1000$ fixed and vary the recorded Fourier dictionary dimension over $J_n \in \{140, 160, \dots, 300\}$. For a fitted curve $\hat{f}$, the reported design-point error is

$$\text{MSE}(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \{\hat{f}(x_i) - f_0(x_i)\}^2.$$

Table 1: Reported design-point MSE, multiplied by 10^3 , across Fourier dictionary dimensions. Boldface marks the lower value within each method pair.

f	Method	Dictionary dimension J_n								
		140	160	180	200	220	240	260	280	300
f_1	Horseshoe	0.21	0.21	0.22	0.19	0.20	0.20	0.28	0.19	0.19
	Ridge	3.82	4.05	4.93	5.42	5.98	6.41	6.95	7.41	8.12
f_2	Horseshoe	0.65	0.68	0.67	0.69	0.69	0.71	0.72	0.72	0.74
	Ridge	1.20	1.35	1.50	1.63	1.78	1.90	2.05	2.18	2.38
f_3	Horseshoe	0.51	0.52	0.52	0.53	0.53	0.54	0.54	0.55	0.56
	Ridge	0.63	0.68	0.73	0.77	0.82	0.86	0.92	0.96	1.04

Across every displayed J_n , the Horseshoe fit has lower reported MSE than ridge. Since both methods use the same Fourier dictionary, Table 1 compares regularization schemes, not the approximation power of different bases. The Horseshoe MSE ranges from 0.19 to 0.28, from 0.65 to 0.74, and from 0.51 to 0.56 in the table’s units for f_1, f_2, f_3 , respectively. This is evidence of limited sensitivity over the tested range. It is not a claim of robustness to parameter overdimensioning, because all displayed values satisfy $J_n < n$, nor is it a statement of statistical significance because Monte Carlo uncertainty is not reported in the table.

3.2 Empirical scaling of point estimators

The scaling experiment uses the same three targets, $J_n = \lfloor n^{0.7} \rfloor$, and the common sample-size grid $n = 1000, 1500, \dots, 20000$. For each available replication, the coefficientwise posterior mean and coefficientwise posterior median are reconstructed as functions, and their errors are measured by

$$\text{RMSE}(\hat{f}) = \|\hat{f} - f_0\|_{2,n}.$$

At each n , the replication-wise RMSE values are summarized by their median. Posterior inference uses the hierarchical Horseshoe with a shared half-Cauchy global scale and treats σ^2 as known.

Two numerical dictionaries are considered: the cosine system $1, \cos(\pi x), \dots, \cos(\pi J_n x)$ and a single-resolution cubic B-spline basis. Thus the cosine design has $p_n = J_n + 1$ columns, while the spline design has $p_n = J_n$ columns. Neither is the exact dictionary appearing in the corresponding corollary: the cosine system is not the periodic real Fourier dictionary used there, while the ordinary cubic spline basis is not multiresolution and does not provide the order-5 approximation profile required to invoke the corollary for f_3 . These experiments therefore compare two practical basis implementations rather than instantiate Corollary 1.

3.2.1 Rate-scale comparison

Equation (4) and Remark 2 give the formal posterior scale

$$\epsilon_n \asymp \sqrt{\log n} n^{(\gamma-1)/2}$$

when $J_n \asymp n^b$, $a_n \asymp s_n \asymp n^\gamma$, the approximation term does not dominate, and all other assumptions of Theorem 1 hold. This formula motivates a descriptive comparison, but it is not a frequentist risk theorem for the posterior mean or coefficientwise posterior median. Deriving such a result from posterior contraction would require additional posterior-moment or uniform-integrability control.

For each function, basis, and point estimator, we regress $\log(\text{RMSE})$ on $\log n$. We also report the slope after replacing the response by

$$\log(\text{RMSE}) - \frac{1}{2} \log \log n.$$

The second regression examines the particular $\sqrt{\log n}$ factor in the polynomial choice of Remark 2; it does not remove the logarithmic factor in the sharper log-balanced rate of Corollary 1, whose exponent is $\alpha/(2\alpha + 1)$. The pure polynomial exponents associated with the one-dimensional minimax benchmark are $-\alpha/(2\alpha + 1)$, namely $-1/3$, $-3/7$, and $-5/11$ for nominal orders 1, 3, 5.

3.2.2 Thresholded coefficient counts

For replication r , define the descriptive coefficient count

$$\hat{q}_{n,r} = \sum_{j=1}^{p_n} \mathbf{1} \left\{ |\hat{\theta}_{j,n,r}^{\text{mean}}| > \frac{1}{J_n} \right\}.$$

We take its median across the available replications and denote by $\hat{\gamma}_q$ the slope from regressing $\log \hat{q}_n$ on $\log n$. This quantity is not the a_n of Assumption 2: the latter is a deterministic sparsity bound for an approximating vector, whereas $\hat{q}_n$ is computed from an estimated posterior summary. The threshold also differs from the corollary threshold $\delta_n = c_\delta \tilde{\epsilon}_n / J_n = o(1/J_n)$. In addition, absolute coefficient thresholds are sensitive to basis scaling; the cosine and localized B-spline columns used here are not normalized to make coefficient magnitudes directly comparable. We therefore report $\hat{\gamma}_q$ only as a basis-specific diagnostic and do not insert it into Theorem 1 to predict a contraction rate.

Table 2 shows a consistent descriptive pattern within this implementation: the cosine fits have more negative RMSE slopes than the single-resolution cubic B-spline fits. The logarithmic adjustment simply makes every fitted slope more negative; it does not, by itself,

Table 2: Empirical log–log RMSE slopes on the common grid $n = 1000, 1500, \dots, 20000$. The adjusted column uses the response $\log(\text{RMSE}) - \frac{1}{2} \log \log n$. The benchmark column records only the pure polynomial exponent $-\alpha/(2\alpha + 1)$. The last column is the exponent of the descriptive $1/J_n$ coefficient count and is not the theoretical approximation sparsity a_n .

f	Basis	Estimator	Raw	Adjusted	Benchmark	$\hat{\gamma}_q$
f_1 , order 1	B-spline	Mean	−0.151	−0.210	−0.333	0.707
		Median	−0.156	−0.214		
	Cosine	Mean	−0.332	−0.390		0.141
		Median	−0.291	−0.349		
f_2 , order 3	B-spline	Mean	−0.145	−0.203	−0.429	0.718
		Median	−0.149	−0.208		
	Cosine	Mean	−0.339	−0.397		0.251
		Median	−0.322	−0.380		
f_3 , order 5	B-spline	Mean	−0.124	−0.182	−0.455	0.747
		Median	−0.138	−0.196		
	Cosine	Mean	−0.327	−0.385		0.325
		Median	−0.332	−0.390		

establish the presence of a $\sqrt{\log n}$ factor. The threshold-count exponents are larger for the B-spline fits, but the basis-normalization and threshold differences prevent a theoretical or causal interpretation of that comparison. In particular, none of these observations verifies the product-prior conditions, the sparse approximants, or the posterior tail conclusion of Theorem 1.

3.3 Real-data illustrations

We finally use the large-dictionary Horseshoe as a descriptive smoother for two time series: daily S&P 500 closing prices from 2015 to 2024 and annual Global Mean Sea Level (GMSL) anomalies from 1880 to 2019. These examples are not applications of Theorem 1. In particular, serial dependence, heteroskedasticity, and unknown noise levels are plausible in both data sets, whereas the theorem assumes independent Gaussian errors with known common variance.

For the S&P 500 series, let y_i be the observed closing price on day i and scale the time index to $x_i \in [0, 1]$. We use a Fourier expansion with $J_n = \lfloor n/2 \rfloor$ as a practical large fixed dictionary. This value is a computational choice, not one prescribed by Corollary 1, and the observed endpoints are not assumed to satisfy the corollary’s periodicity condition. Figure 3 compares the resulting smooth with the observed price series.

For the GMSL series (Liu and Li, 2025), we use a cubic B-spline basis with uniformly spaced

interior knots and $J_n = \lfloor n/2 \rfloor$. The fit again uses the hierarchical Horseshoe with a shared random global scale. Thus both the prior and the single-resolution spline dictionary differ from those covered by Theorem 1 and the spline corollaries.

4 Conclusion

This paper introduces a Bayesian nonparametric regression approach that addresses the trade-off between theoretical optimality and computational feasibility. Employing a large deterministic number of basis functions scaling as n^b with shrinkage priors provides a statistically principled route to adaptive posterior contraction. When the dictionary growth and, for splines, the approximation order are fixed before the truth, one deterministic product-Horseshoe prior is pointwise adaptive throughout the admissible smoothness range described in Remark 1; the result does not assert uniform contraction over the smoothness index.

Under the nested sparse-approximation and iid product-prior conditions stated in the corollaries, the posterior distribution achieves the minimax-optimal contraction rate of $n^{-\alpha/(2\alpha+1)}$ up to logarithmic factors for functions in $C^\alpha[0, 1]$. This conclusion extends to the multivariate case with rates of $n^{-\alpha/(2\alpha+d)}$ for $C^\alpha([0, 1]^d)$, and to anisotropic smoothness with rates governed by the harmonic mean of directional smoothness parameters.

The numerical examples demonstrate the feasibility of large fixed dictionaries with practical Horseshoe regularization. Because those computations use a shared random global scale and ordinary numerical dictionaries, their point-estimator performance is descriptive and is not presented as an empirical verification of the product-prior contraction theorem.

5 Appendix

The deterministic-scale Horseshoe prior

The deterministic-scale Horseshoe specialization (Carvalho et al., 2009) covered by Theorem 1 is the product prior obtained from

$$\theta_j \mid \lambda_j, \tau_n \sim N(0, \tau_n^2 \lambda_j^2), \quad \lambda_j \stackrel{\text{iid}}{\sim} C^+(0, 1), \quad j = 1, \dots, J_n,$$

where the global scale τ_n is deterministic. After the independent local scales are integrated out, the coefficient prior remains the iid product $\bigotimes_{j=1}^{J_n} \pi_{\tau_n}$. This distinction is essential: a shared random global scale makes the coordinates dependent after marginalization and is not covered by Theorem 1.

Under the substitutions $p = J_n$ and $s_* = a_n$, [Wei and Ghosal \(2020, Theorem 3.3\)](#) also assume a bounded active truth and establish their posterior contraction property (9) at rate $\sqrt{s_* \log p/n}$. To imply (WG3) with the constant μ fixed in Assumption 3, take a constant $\mu'' > \mu$; the relevant scale and signal-size conditions in that theorem are

$$\tau_n < \frac{J_n^{-(2+\mu'')}}{\sqrt{n}}, \quad \max\{\log E_n, -\log \tau_n\} = O(\log J_n). \quad (29)$$

Its proof verifies conditions (3)–(5) for a threshold below $\tilde{\epsilon}_n/J_n$. The coefficient-box condition (6) used in that paper is not needed for the shell-covering argument below.

For completeness, a direct sufficient specialization of Assumption 3 at $\delta_n = c_\delta \tilde{\epsilon}_n/J_n$ is obtained by taking $J_n \asymp n^b$ for a fixed $b > 0$, $E_n \leq J_n^C$ for a fixed $C > 0$, $a_n \log J_n \lesssim n\tilde{\epsilon}_n^2$, and

$$\tau_n = \frac{J_n^{-(2+\kappa+\rho)}}{\sqrt{n}} \quad (30)$$

for fixed $c_\delta \in (0, 1)$, $\kappa > \mu$, and $\rho > 0$. To see this, write $\theta_j = \tau_n \lambda_j Z_j$, where $Z_j \sim N(0, 1)$ is independent of λ_j , and put $T_n = J_n^{1+\kappa}$. The half-Cauchy and Gaussian tail bounds give, for every $t > 0$,

$$\begin{aligned} \Pi_{n,j}(|\theta_j| > t) &\leq \Pr(\lambda_j > T_n) + \Pr\left(|Z_j| > \frac{t}{\tau_n T_n}\right) \\ &\leq \frac{2}{\pi T_n} + 2 \exp\left(-\frac{t^2}{2\tau_n^2 T_n^2}\right). \end{aligned}$$

At $t = \delta_n$ this becomes

$$\begin{aligned} \Pi_{n,j}(|\theta_j| > \delta_n) &\leq \frac{2}{\pi} J_n^{-(1+\kappa)} + 2 \exp\left(-\frac{c_\delta^2}{2} n \tilde{\epsilon}_n^2 J_n^{2\rho}\right) \\ &\leq J_n^{-(1+\mu)} \end{aligned}$$

eventually, which verifies (WG3). To verify (WG4), let $m_n = J_n - a_n$. If $m_n = 0$, the condition is automatic. Otherwise put $t_n = \eta \tilde{\epsilon}_n/m_n \geq \eta \tilde{\epsilon}_n/J_n$. The same tail bound gives

$$q_n := \Pi_{n,j}(|\theta_j| > t_n) \leq \frac{2}{\pi} J_n^{-(1+\kappa)} + 2 \exp\left(-\frac{\eta^2}{2} n \tilde{\epsilon}_n^2 J_n^{2\rho}\right) = o(J_n^{-1}).$$

Independence therefore yields

$$\Pi_n\left(\sum_{j \notin \xi_n^*} |\theta_j| \leq \eta \tilde{\epsilon}_n\right) \geq \Pi_n(|\theta_j| \leq t_n \text{ for every } j \notin \xi_n^*)$$

$$= (1 - q_n)^{m_n} \geq 1 - m_n q_n \longrightarrow 1.$$

Thus the negative logarithm in (WG4) is $o(1)$ and is bounded by $c_4 n \tilde{\epsilon}_n^2$ for a fixed $c_4 > 0$ and all sufficiently large n .

Finally, the scaled marginal density obeys

$$\pi_{\tau_n}(t) = \tau_n^{-1} \pi_1(t/\tau_n), \quad \pi_1(u) \geq \frac{c}{1 + u^2}$$

for a universal constant $c > 0$. Indeed, the normal–half-Cauchy scale-mixture formula is

$$\pi_1(u) = \frac{2}{\pi\sqrt{2\pi}} \int_0^\infty \frac{\exp\{-u^2/(2\lambda^2)\}}{\lambda(1 + \lambda^2)} d\lambda.$$

Integrating over $\lambda \in [1, 2]$ when $|u| \leq 1$ and over $\lambda \in [|u|, 2|u|]$ when $|u| > 1$ gives the displayed lower bound. Consequently,

$$\pi_{n,E_n} \geq \frac{c\tau_n}{E_n^2 + \tau_n^2}.$$

Since $J_n \asymp n^b$, $E_n \leq J_n^C$, and τ_n is polynomial in (J_n, n) , one has $-\log \pi_{n,E_n} = O(\log J_n)$. Also, $\log a_n \leq \log J_n$ and $\log(1/\tilde{\epsilon}_n) = O(\log J_n)$ because $n\tilde{\epsilon}_n^2 \rightarrow \infty$. Hence $a_n \log J_n \lesssim n\tilde{\epsilon}_n^2$ verifies (WG5).

Main theorem

Proof of Theorem 1. Define the sieve

$$W_n = \left\{ \theta \in \mathbb{R}^{J_n} : \#\{j : |\theta_j| > \delta_n\} \leq A s_n \right\},$$

with $A \geq 1$ a fixed integer chosen below. The constants are fixed in the following order, each depending only on quantities fixed before it:

$$\tilde{L} = 1 + K(1 + 2\eta), \quad C_0 = \frac{\sqrt{2}\tilde{L}}{\sigma} + \frac{\tilde{L}^2}{2\sigma^2}, \quad C_1 = c_4 + c_5 + C_0, \quad A = \left\lceil \frac{2(C_1 + 1)}{\mu} \right\rceil,$$

$$M = \max \left\{ 16K, 4\sigma\sqrt{2A + 1}, 8\sigma\sqrt{C_1 + 1} \right\}.$$

There is no circularity: $\tilde{L}$ depends only on (K, η) ; C_0 on $(\tilde{L}, \sigma)$; C_1 on (C_0, c_4, c_5) ; A on (C_1, μ) ; and M on (K, σ, A, C_1) .

Write

$$D_n = \int_{\mathbb{R}^{J_n}} \frac{p_\theta}{p_0}(Y^{(n)}) d\Pi_n(\theta)$$

for the likelihood-ratio evidence, which is the denominator in Bayes' formula after division by $p_0(Y^{(n)})$.

The argument constructs tests on the sieve, controls the prior mass outside the sieve, and lower-bounds the evidence before combining these estimates. The prior concentration rate $\tilde{\epsilon}_n$ controls the local prior mass and the evidence, whereas the contraction rate ϵ_n controls the sieve entropy, the tests, and the radius of the posterior contraction statement. We begin with the test construction.

Lemma 3 (Basic Gaussian test). Let $g \in \mathbb{R}^n$ with $d = \|g - f_0\|_{2,n} > 0$, and define the test

$$\psi_g = \mathbf{1} \left\{ \sum_{i=1}^n (y_i - f_0(x_i))(g_i - f_0(x_i)) \geq \frac{nd^2}{2} \right\}.$$

Then

$$E_0 \psi_g \leq \exp\left(-\frac{nd^2}{8\sigma^2}\right) \quad \text{and} \quad \sup_{v \in \mathbb{R}^n: \|v - g\|_{2,n} \leq d/4} E_v(1 - \psi_g) \leq \exp\left(-\frac{nd^2}{32\sigma^2}\right).$$

Proof. Write $T = \sum_i (y_i - f_0(x_i))(g_i - f_0(x_i))$ and note $\sum_i (g_i - f_0(x_i))^2 = nd^2$. Under P_0 , $y_i - f_0(x_i) = \varepsilon_i$ with $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$, so $T \sim N(0, \sigma^2 nd^2)$ and, by the Gaussian tail bound $\Pr(Z \geq t) \leq e^{-t^2/2}$ for $Z \sim N(0, 1)$ and $t \geq 0$,

$$E_0 \psi_g = \Pr\left(Z \geq \frac{\sqrt{n} d}{2\sigma}\right) \leq \exp\left(-\frac{nd^2}{8\sigma^2}\right).$$

Under P_v with $\|v - g\|_{2,n} \leq d/4$, $T \sim N(m, \sigma^2 nd^2)$ with mean

$$\begin{aligned} m &= \sum_{i=1}^n (v_i - f_0(x_i))(g_i - f_0(x_i)) \\ &= n \langle v - f_0, g - f_0 \rangle_n \\ &= nd^2 + n \langle v - g, g - f_0 \rangle_n \\ &\geq nd^2 - n \|v - g\|_{2,n} d \geq \frac{3nd^2}{4}, \end{aligned}$$

using the Cauchy–Schwarz inequality. Hence

$$E_v(1 - \psi_g) = P_v\left(T < \frac{nd^2}{2}\right)$$

$$\begin{aligned}
&\leq \Pr \left(N(0, \sigma^2 n d^2) < -\frac{n d^2}{4} \right) \\
&\leq \exp \left(-\frac{(n d^2/4)^2}{2 \sigma^2 n d^2} \right) = \exp \left(-\frac{n d^2}{32 \sigma^2} \right).
\end{aligned}$$

□

For each integer $\ell \geq 1$ define the shell

$$\mathcal{S}_\ell = \left\{ \theta \in W_n : \ell M \epsilon_n < \|f_\theta - f_0\|_{2,n} \leq (\ell + 1) M \epsilon_n \right\},$$

so that $\{\theta \in W_n : \|f_\theta - f_0\|_{2,n} > M \epsilon_n\} = \bigcup_{\ell \geq 1} \mathcal{S}_\ell$.

Lemma 4 (Covering of the shells). For every $\ell \geq 1$ there exist N_ℓ points $g_{\ell,1}, \dots, g_{\ell,N_\ell} \in \mathbb{R}^n$ such that

$$\|g_{\ell,k} - f_0\|_{2,n} \geq \frac{15}{16} \ell M \epsilon_n \quad \text{for every } k,$$

every $\theta \in \mathcal{S}_\ell$ satisfies $\|f_\theta - g_{\ell,k}\|_{2,n} \leq \ell M \epsilon_n / 8 \leq \|g_{\ell,k} - f_0\|_{2,n} / 4$ for some k , and, for all large n uniformly in ℓ ,

$$N_\ell \leq \exp\{2A s_n \log J_n\} = \exp\{2A n \epsilon_n^2\}.$$

Proof. Fix $\ell \geq 1$. For $\theta \in W_n$ let $S(\theta) = \{j : |\theta_j| > \delta_n\}$, so $|S(\theta)| \leq A s_n$, and decompose $\theta = \theta^{(1)} + \theta^{(2)}$, where $\theta^{(1)}$ retains the coordinates in $S(\theta)$ and $\theta^{(2)}$ the rest. By Assumption 1, the triangle inequality, and $\delta_n \leq \tilde{\epsilon}_n / J_n \leq \epsilon_n / J_n$,

$$\|f_{\theta^{(2)}}\|_{2,n} \leq \sum_{j \notin S(\theta)} |\theta_j| \|\xi_j\|_{2,n} \leq K J_n \delta_n \leq K \epsilon_n \leq \frac{M \epsilon_n}{16},$$

since $M \geq 16K$. Consequently, for $\theta \in \mathcal{S}_\ell$,

$$\begin{aligned}
\|f_{\theta^{(1)}} - f_0\|_{2,n} &\leq \|f_\theta - f_0\|_{2,n} + \|f_{\theta^{(2)}}\|_{2,n} \\
&\leq (\ell + 1) M \epsilon_n + \frac{M \epsilon_n}{16} = \left(\ell + \frac{17}{16} \right) M \epsilon_n \\
&\leq (\ell + 2) M \epsilon_n,
\end{aligned}$$

and, in the other direction,

$$\begin{aligned}
\|f_{\theta^{(1)}} - f_0\|_{2,n} &\geq \|f_\theta - f_0\|_{2,n} - \|f_{\theta^{(2)}}\|_{2,n} \\
&> \ell M \epsilon_n - \frac{M \epsilon_n}{16} = \left(\ell - \frac{1}{16} \right) M \epsilon_n
\end{aligned}$$

$$\geq \frac{15}{16} \ell M \epsilon_n,$$

where the last inequality uses $\ell \geq 1$.

Fix a support $S \subset \{1, \dots, J_n\}$ with $|S| = s \leq As_n$ and let $V_S \subset \mathbb{R}^n$ be the linear span of the evaluation vectors $\{\xi_j : j \in S\}$, a subspace of dimension at most s . Consider the set

$$\mathcal{F}_{\ell,S} = \{f_{\theta^{(1)}} : \theta \in \mathcal{S}_\ell, S(\theta) = S\} \subset V_S \cap \{v : \|v - f_0\|_{2,n} \leq (\ell + 2)M\epsilon_n\}.$$

Let $r = \ell M \epsilon_n / 16$. If $\mathcal{F}_{\ell,S} = \emptyset$, set $N_{\ell,S} = 0$. Otherwise choose, deterministically, a finite maximal r -separated subset $\{g_{\ell,S,1}, \dots, g_{\ell,S,N_{\ell,S}}\}$ of $\mathcal{F}_{\ell,S}$ with respect to $\|\cdot\|_{2,n}$. Such a finite set exists because $\mathcal{F}_{\ell,S}$ is bounded in a finite-dimensional normed space and hence is totally bounded. Maximality implies that every element of $\mathcal{F}_{\ell,S}$ lies within distance r of some $g_{\ell,S,k}$, and each $g_{\ell,S,k}$ belongs to $\mathcal{F}_{\ell,S}$, hence satisfies $\|g_{\ell,S,k} - f_0\|_{2,n} \geq \frac{15}{16} \ell M \epsilon_n$. To count the points, let P denote the orthogonal projection onto V_S with respect to $\langle \cdot, \cdot \rangle_n$; by the Pythagorean identity, every $v \in V_S$ with $\|v - f_0\|_{2,n} \leq R$, $R = (\ell + 2)M\epsilon_n$, satisfies $\|v - Pf_0\|_{2,n} \leq R$, so $\mathcal{F}_{\ell,S}$ is contained in the ball of radius R around Pf_0 inside V_S . The balls of radius $r/2$ around the separated points are disjoint and contained in the ball of radius $R + r/2$ around Pf_0 inside V_S ; comparing volumes in the at most s -dimensional Euclidean space $(V_S, \|\cdot\|_{2,n})$,

$$N_{\ell,S} \leq \left(\frac{R + r/2}{r/2} \right)^s = \left(\frac{2R}{r} + 1 \right)^s = \left(\frac{32(\ell + 2)}{\ell} + 1 \right)^s \leq 97^s \leq 97^{As_n},$$

using $(\ell + 2)/\ell \leq 3$ for $\ell \geq 1$. Taking the union over all supports S of size at most $q_n = \min\{As_n, J_n\}$ yields the collection $\{g_{\ell,k}\}$ with

$$N_\ell \leq \sum_{s=0}^{q_n} \binom{J_n}{s} 97^{As_n} \leq (As_n + 1) J_n^{As_n} 97^{As_n},$$

so that, whenever $N_\ell \geq 1$,

$$\log N_\ell \leq As_n \log J_n + As_n \log 97 + \log(As_n + 1) \leq 2As_n \log J_n$$

for all large n , uniformly in ℓ , because $J_n \rightarrow \infty$ makes $\log 97 \leq \frac{1}{2} \log J_n$ and $\log(As_n + 1) \leq \frac{1}{2} As_n \log J_n$ eventually. The claimed bound on N_ℓ also holds when $N_\ell = 0$. Finally, for $\theta \in \mathcal{S}_\ell$ with $S(\theta) = S$ and its assigned center $g = g_{\ell,S,k}$,

$$\begin{aligned} \|f_\theta - g\|_{2,n} &\leq \|f_{\theta^{(1)}} - g\|_{2,n} + \|f_{\theta^{(2)}}\|_{2,n} \\ &\leq \frac{\ell M \epsilon_n}{16} + \frac{M \epsilon_n}{16} \leq \frac{\ell M \epsilon_n}{8} \\ &\leq \frac{1}{4} \cdot \frac{15}{16} \ell M \epsilon_n \leq \frac{\|g - f_0\|_{2,n}}{4}, \end{aligned}$$

since $\ell + 1 \leq 2\ell$ and $1/8 \leq 15/64$. This proves the lemma. $\square$

Define the global test

$$\phi_n = \sup_{\ell \geq 1} \sup_{1 \leq k \leq N_\ell} \psi_{g_{\ell,k}}, \quad \sup \emptyset := 0.$$

This is measurable because it is a countable supremum of Borel half-space tests. For each fixed truth f_0 , it is a Borel measurable function of the data, but both the centers and the tests depend on f_0 ; as is standard in pointwise posterior-contraction arguments, they are analytic testing devices and are not proposed as implementable procedures.

For the Type I error, Lemmas 3 and 4, together with $\|g_{\ell,k} - f_0\|_{2,n} \geq \frac{15}{16}\ell M\epsilon_n$, give

$$\frac{n \|g_{\ell,k} - f_0\|_{2,n}^2}{8\sigma^2} \geq \frac{225}{256} \cdot \frac{\ell^2 M^2 n \epsilon_n^2}{8\sigma^2} \geq \frac{\ell^2 M^2 n \epsilon_n^2}{16\sigma^2},$$

Consequently, with $c = M^2 n \epsilon_n^2 / (16\sigma^2) \geq 1$ for all large n (because $n \epsilon_n^2 \rightarrow \infty$),

$$\begin{aligned} E_0 \phi_n &\leq \sum_{\ell \geq 1} N_\ell e^{-\ell^2 c} \leq e^{2A n \epsilon_n^2} \sum_{\ell \geq 1} e^{-\ell c} \\ &= e^{2A n \epsilon_n^2} \frac{e^{-c}}{1 - e^{-c}} \\ &\leq 2 \exp \left(2A n \epsilon_n^2 - \frac{M^2 n \epsilon_n^2}{16\sigma^2} \right) \leq 2e^{-n \epsilon_n^2}, \end{aligned}$$

where the last inequality uses $M^2 \geq 16\sigma^2(2A + 1)$, guaranteed by $M \geq 4\sigma\sqrt{2A + 1}$.

For the Type II error, fix $\ell \geq 1$ and $\theta \in \mathcal{S}_\ell$, and let $g = g_{\ell,k}$ be its assigned center, so $\|f_\theta - g\|_{2,n} \leq \|g - f_0\|_{2,n}/4$. By Lemma 3 applied at $v = f_\theta$,

$$\begin{aligned} E_{f_\theta}(1 - \phi_n) &\leq E_{f_\theta}(1 - \psi_g) \leq \exp \left(-\frac{n \|g - f_0\|_{2,n}^2}{32\sigma^2} \right) \\ &\leq \exp \left(-\frac{225}{256} \cdot \frac{\ell^2 M^2 n \epsilon_n^2}{32\sigma^2} \right) \\ &\leq \exp \left(-\frac{\ell^2 M^2 n \epsilon_n^2}{64\sigma^2} \right). \end{aligned}$$

If $\theta \in W_n$ instead satisfies $\|f_\theta - f_0\|_{2,n} > \ell M\epsilon_n$, then $\theta \in \mathcal{S}_m$ for some $m \geq \ell$; the bound just proved with $m^2 \geq \ell^2$ gives the required uniform Type II bound. Together with the Type I bound, this completes the test construction.

We next control the prior mass outside the sieve.

Let $N = \#\{j : |\theta_j| > \delta_n\}$. Under Π_n the coordinates are independent with common exceedance probability

$$\nu_n = \int_{|t| > \delta_n} \pi_n(t) dt \leq J_n^{-(1+\mu)}$$

by condition (WG3), so under Π_n the variable N is Binomial(J_n, ν_n). If $As_n \geq J_n$, then $W_n^c = \emptyset$ and the desired sieve-complement bound is immediate. Suppose henceforth that $As_n < J_n$. For any integer $k \geq 1$, the union bound over the $\binom{J_n}{k}$ possible sets of k exceeding indices gives $\Pi_n(N \geq k) \leq \binom{J_n}{k} \nu_n^k$. Since As_n is a positive integer and $\binom{J_n}{k} \leq (eJ_n/k)^k$,

$$\begin{aligned} \Pi_n(W_n^c) &= \Pi_n(N > As_n) \leq \Pi_n(N \geq As_n) \\ &\leq \binom{J_n}{As_n} \nu_n^{As_n} \\ &\leq \left(\frac{eJ_n}{As_n}\right)^{As_n} J_n^{-(1+\mu)As_n} \\ &\leq \exp\{As_n - \mu As_n \log J_n\}, \end{aligned}$$

where $\log(eJ_n/(As_n)) \leq 1 + \log J_n$ was used. Hence

$$\begin{aligned} \Pi_n(W_n^c) &\leq \exp(-\mu As_n \log J_n + As_n) \\ &\leq \exp\left(-\frac{\mu A}{2} s_n \log J_n\right) = \exp\left(-\frac{\mu A}{2} n \epsilon_n^2\right) \end{aligned}$$

for all large n , since $As_n \leq \frac{\mu}{2} As_n \log J_n$ once $\log J_n \geq 2/\mu$. This is the required sieve-complement estimate.

We next construct a neighborhood of f_0 with sufficient prior mass and use it to lower-bound the evidence.

We begin with the prior mass of an empirical ball around f_0 . Let θ_0 and ξ_n^* be as in Assumption 2. For every $j \in \xi_n^*$ define an inward one-sided interval of length $\eta \tilde{\epsilon}_n/a_n$ by

$$I_j = \begin{cases} [\theta_{0,j} - \eta \tilde{\epsilon}_n/a_n, \theta_{0,j}], & \theta_{0,j} > 0, \\ [\theta_{0,j}, \theta_{0,j} + \eta \tilde{\epsilon}_n/a_n], & \theta_{0,j} \leq 0. \end{cases}$$

Because $|\theta_{0,j}| \leq E_n$, E_n is positive and nondecreasing, and $\eta \tilde{\epsilon}_n/a_n \rightarrow 0$, these intervals are contained in $[-E_n, E_n]$ for all large n . Define

$$G_n = \{\theta_j \in I_j \text{ for every } j \in \xi_n^*\} \cap \left\{ \sum_{j \notin \xi_n^*} |\theta_j| \leq \eta \tilde{\epsilon}_n \right\}.$$

On G_n , using $|\xi_n^*| = a_n$, $|\theta_{0,j}| \leq \delta_n$ for $j \notin \xi_n^*$, and $J_n \delta_n \leq \tilde{\epsilon}_n$,

$$\begin{aligned} \|\theta - \theta_0\|_1 &= \sum_{j \in \xi_n^*} |\theta_j - \theta_{0,j}| + \sum_{j \notin \xi_n^*} |\theta_j - \theta_{0,j}| \\ &\leq a_n \frac{\eta \tilde{\epsilon}_n}{a_n} + \sum_{j \notin \xi_n^*} |\theta_j| + \sum_{j \notin \xi_n^*} |\theta_{0,j}| \\ &\leq \eta \tilde{\epsilon}_n + \eta \tilde{\epsilon}_n + J_n \delta_n \leq (1 + 2\eta) \tilde{\epsilon}_n. \end{aligned}$$

Consequently, by Assumptions 1 and 2,

$$\begin{aligned} \|f_\theta - f_0\|_{2,n} &\leq \|f_0 - f_{\theta_0}\|_{2,n} + \sum_{j=1}^{J_n} |\theta_j - \theta_{0,j}| \|\xi_j\|_{2,n} \\ &\leq \tilde{\epsilon}_n + K(1 + 2\eta) \tilde{\epsilon}_n = \tilde{L} \tilde{\epsilon}_n. \end{aligned}$$

Thus $G_n \subset B_n$, where

$$B_n = \left\{ \theta : \|f_\theta - f_0\|_{2,n} \leq \tilde{L} \tilde{\epsilon}_n \right\}.$$

For every active coordinate, the density floor in (WG5) and the fact that $I_j \subset [-E_n, E_n]$ give

$$0 < b_n := \frac{\eta \tilde{\epsilon}_n}{a_n} \pi_{n,E_n} \leq \Pi_{n,j}(\theta_j \in I_j) \leq 1.$$

Since $|\xi_n^*| = a_n$, the active block has probability at least $b_n^{a_n}$. Moreover, because $\tilde{\epsilon}_n \rightarrow 0$, eventually $\tilde{\epsilon}_n \leq \eta$, and hence

$$\left[\log \left(\frac{a_n}{\eta \tilde{\epsilon}_n \pi_{n,E_n}} \right) \right]_+ \leq \left[\log a_n + 2 \log \left(\frac{1}{\tilde{\epsilon}_n} \right) - \log \pi_{n,E_n} \right]_+.$$

It follows from (WG5) that the active block has probability at least $e^{-c_5 n \tilde{\epsilon}_n^2}$. The active and inactive blocks are independent, and (WG4) gives the inactive-block probability at least $e^{-c_4 n \tilde{\epsilon}_n^2}$. Therefore

$$\Pi_n(B_n) \geq \Pi_n(G_n) \geq e^{-(c_4 + c_5) n \tilde{\epsilon}_n^2}.$$

We now derive the evidence lower bound. Under P_0 , write $Z_i = \varepsilon_i / \sigma$, so that $Z_1, \dots, Z_n$ are independent $N(0, 1)$ variables. Then, for every θ ,

$$\log \frac{p_\theta}{p_0}(Y^{(n)}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n \left[(y_i - f_\theta(x_i))^2 - (y_i - f_0(x_i))^2 \right]$$

$$= \frac{1}{\sigma} \sum_{i=1}^n Z_i (f_\theta - f_0)(x_i) - \frac{n}{2\sigma^2} \|f_\theta - f_0\|_{2,n}^2.$$

In particular the Kullback–Leibler divergence and the centered second moment are exactly

$$E_0 \left[\log \frac{p_0}{p_\theta} \right] = \frac{n}{2\sigma^2} \|f_\theta - f_0\|_{2,n}^2, \quad \text{Var}_0 \left(\log \frac{p_0}{p_\theta} \right) = \frac{n}{\sigma^2} \|f_\theta - f_0\|_{2,n}^2,$$

and hence B_n is contained in a centered Kullback–Leibler neighborhood of p_0 whose divergence and variance radii are constant multiples of $n\tilde{\epsilon}_n^2$.

Let $\Pi_{n,B_n}(\cdot) = \Pi_n(\cdot \cap B_n)/\Pi_n(B_n)$ denote the renormalized prior. By restriction to B_n and Jensen’s inequality applied to the convex function $-\log$,

$$\begin{aligned} \log \frac{D_n}{\Pi_n(B_n)} &\geq \log \int \frac{p_\theta}{p_0}(Y^{(n)}) d\Pi_{n,B_n}(\theta) \\ &\geq \int \log \frac{p_\theta}{p_0}(Y^{(n)}) d\Pi_{n,B_n}(\theta) \\ &= \frac{1}{\sigma} \sum_{i=1}^n Z_i \bar{g}(x_i) - \frac{n}{2\sigma^2} \int \|f_\theta - f_0\|_{2,n}^2 d\Pi_{n,B_n}(\theta), \end{aligned}$$

where $\bar{g} = \int (f_\theta - f_0) d\Pi_{n,B_n}(\theta)$, the integral taken coordinatewise on evaluation vectors; interchanging the sum and the integral is justified because on B_n the integrand is bounded for each fixed $Y^{(n)}$. On B_n , $\|f_\theta - f_0\|_{2,n} \leq \tilde{L}\tilde{\epsilon}_n$, so the quadratic term is bounded below by $-n\tilde{L}^2\tilde{\epsilon}_n^2/(2\sigma^2)$, and by convexity of the norm and Jensen’s inequality,

$$\|\bar{g}\|_{2,n} \leq \int \|f_\theta - f_0\|_{2,n} d\Pi_{n,B_n}(\theta) \leq \tilde{L}\tilde{\epsilon}_n.$$

The linear term $Z = \sigma^{-1} \sum_i Z_i \bar{g}(x_i)$ is, under P_0 , a centered Gaussian with variance $v = n \|\bar{g}\|_{2,n}^2 / \sigma^2 \leq n\tilde{L}^2\tilde{\epsilon}_n^2 / \sigma^2$. If $v = 0$ then $Z = 0$ almost surely; otherwise, by the Gaussian tail bound with $t = \sqrt{2v n\tilde{\epsilon}_n^2}$,

$$P_0 \left(Z \leq -\sqrt{2v n\tilde{\epsilon}_n^2} \right) \leq \exp \left(-\frac{2v n\tilde{\epsilon}_n^2}{2v} \right) = e^{-n\tilde{\epsilon}_n^2}, \quad \sqrt{2v n\tilde{\epsilon}_n^2} \leq \frac{\sqrt{2}\tilde{L}}{\sigma} n\tilde{\epsilon}_n^2.$$

Therefore, with P_0 -probability at least $1 - e^{-n\tilde{\epsilon}_n^2}$,

$$\log D_n \geq \log \Pi_n(B_n) - \frac{\sqrt{2}\tilde{L}}{\sigma} n\tilde{\epsilon}_n^2 - \frac{\tilde{L}^2}{2\sigma^2} n\tilde{\epsilon}_n^2 \geq -(c_4 + c_5 + C_0) n\tilde{\epsilon}_n^2 = -C_1 n\tilde{\epsilon}_n^2,$$

which is the required evidence lower bound.

We finally combine the preceding estimates to obtain the quantitative $L_1(P_0)$ bound and posterior contraction.

Let

$$U_n = \left\{ \theta : \|f_0 - f_\theta\|_{2,n} \leq M\epsilon_n \right\}, \quad Z_n = \Pi_n(U_n^c \mid Y^{(n)}), \quad d_n^* = e^{-C_1 n \tilde{\epsilon}_n^2}.$$

Since $0 \leq Z_n \leq 1$, split its expectation according to the test and the evidence event:

$$\begin{aligned} E_0 Z_n &= E_0(\phi_n Z_n) + E_0[(1 - \phi_n) \mathbf{1}\{D_n < d_n^*\} Z_n] + R_n \\ &\leq E_0 \phi_n + P_0(D_n < d_n^*) + R_n, \end{aligned}$$

where

$$R_n = E_0[(1 - \phi_n) \mathbf{1}\{D_n \geq d_n^*\} Z_n].$$

On $\{D_n \geq d_n^*\}$, split the posterior numerator over W_n^c and W_n to obtain

$$(1 - \phi_n) Z_n \leq \frac{1}{d_n^*} \left[\int_{W_n^c} \frac{p_\theta}{p_0} d\Pi_n(\theta) + \int_{U_n^c \cap W_n} (1 - \phi_n) \frac{p_\theta}{p_0} d\Pi_n(\theta) \right].$$

Tonelli's theorem and the change-of-measure identity

$$E_0 \left[\frac{p_\theta}{p_0}(Y^{(n)}) h(Y^{(n)}) \right] = E_{f_\theta} [h(Y^{(n)})]$$

for every nonnegative measurable h consequently give

$$R_n \leq \frac{1}{d_n^*} \left[\Pi_n(W_n^c) + \int_{U_n^c \cap W_n} E_{f_\theta}(1 - \phi_n) d\Pi_n(\theta) \right].$$

Since $U_n^c \cap W_n = \bigcup_{\ell \geq 1} \mathcal{S}_\ell$, the uniform Type II bound gives, with $c' = M^2 n \epsilon_n^2 / (64\sigma^2) \geq 1$ for all large n ,

$$\begin{aligned} \int_{U_n^c \cap W_n} E_{f_\theta}(1 - \phi_n) d\Pi_n(\theta) &\leq \sum_{\ell \geq 1} \Pi_n(\mathcal{S}_\ell) e^{-\ell^2 c'} \\ &\leq \sum_{\ell \geq 1} e^{-\ell c'} \leq 2 \exp\left(-\frac{M^2 n \epsilon_n^2}{64\sigma^2}\right). \end{aligned}$$

Using the sieve-complement estimate, $\tilde{\epsilon}_n \leq \epsilon_n$, and the definitions of A and M , we first obtain

$$\begin{aligned} R_n &\leq e^{C_1 n \tilde{\epsilon}_n^2} \left[e^{-(\mu A/2) n \epsilon_n^2} + 2e^{-M^2 n \epsilon_n^2 / (64\sigma^2)} \right] \\ &\leq 3e^{-n \epsilon_n^2}. \end{aligned}$$

The Type I and evidence bounds and the expectation decomposition therefore give the quantitative estimate

$$\begin{aligned} E_0 Z_n &\leq 2e^{-n\epsilon_n^2} + e^{-n\tilde{\epsilon}_n^2} + 3e^{-n\epsilon_n^2} \\ &\leq 6e^{-n\tilde{\epsilon}_n^2}. \end{aligned}$$

This proves the asserted $L_1(P_0)$ convergence. Moreover, for every fixed $\zeta \in (0, 1)$, Markov's inequality gives

$$P_0(Z_n > \zeta) \leq \frac{E_0 Z_n}{\zeta} \leq \frac{6}{\zeta} e^{-n\tilde{\epsilon}_n^2} \longrightarrow 0,$$

because $n\tilde{\epsilon}_n^2 \rightarrow \infty$; for $\zeta \geq 1$ the claim is automatic because $0 \leq Z_n \leq 1$. Hence, by the definition of convergence in probability and by the induced-posterior convention introduced above,

$$\Pi_n(\|f_0 - f\|_{2,n} > M\epsilon_n \mid Y^{(n)}) \xrightarrow{P_0} 0.$$

This is posterior contraction at the rate ϵ_n . If $M_n \rightarrow \infty$, then $M_n \geq M$ eventually, and the corresponding posterior tail is bounded by Z_n , proving both stated extensions.

For completeness, let $Q_n = \Pi_n(W_n^c \mid Y^{(n)})$. On $\{D_n \geq d_n^*\}$, its posterior numerator is bounded in the same way as above. Since $0 \leq Q_n \leq 1$, the evidence bound, Tonelli's theorem, the change-of-measure identity with $h = 1$, and the sieve-complement estimate yield

$$\begin{aligned} E_0 Q_n &\leq P_0(D_n < d_n^*) + e^{C_1 n\tilde{\epsilon}_n^2} \Pi_n(W_n^c) \\ &\leq e^{-n\tilde{\epsilon}_n^2} + e^{-n\epsilon_n^2} \leq 2e^{-n\tilde{\epsilon}_n^2}. \end{aligned}$$

Thus, for every fixed $\zeta > 0$,

$$P_0(Q_n > \zeta) \leq \frac{E_0 Q_n}{\zeta} \leq \frac{2}{\zeta} e^{-n\tilde{\epsilon}_n^2} \longrightarrow 0,$$

which proves the effective-support conclusion and its asserted $L_1(P_0)$ strengthening. $\square$

Proofs of Corollaries 1–3

The displayed sparse approximation profiles are the only basis-specific input beyond Assumption 1. Lemmas 1 and 2 verify both ingredients for the concrete Fourier and multiresolution spline dictionaries, with the spline-order restrictions and anisotropic qualification stated there. The proofs below start from the profiles as stated and verify every remaining

hypothesis of Theorem 1.

Proof of Corollary 1. Since $\log J_n \asymp \log n$ and $b \geq (2\alpha+1)^{-1}$, one has $m_n \leq J_n$ and $a_n \leq J_n$ eventually. Apply (15) at $m = m_n$ and denote the resulting vector by θ_0 . For a sufficiently large C_ϵ ,

$$\|f_0 - f_{\theta_0}\|_{2,n} \leq C_{\text{app}} m_n^{-\alpha} \lesssim N_n^{-\alpha/(2\alpha+1)} \leq \tilde{\epsilon}_n.$$

Moreover, θ_0 has at most a_n nonzero entries and $\|\theta_0\|_\infty \leq E_n$. Hence Assumption 2 holds for every positive threshold, in particular for (17).

The identities

$$n\tilde{\epsilon}_n^2 \asymp nN_n^{-2\alpha/(2\alpha+1)} = N_n^{1/(2\alpha+1)} \log J_n \asymp a_n \log J_n$$

show that $\tilde{\epsilon}_n \downarrow 0$, $n\tilde{\epsilon}_n^2 \rightarrow \infty$, and $s_n \asymp a_n$. Since $a_n \log J_n = o(n)$, all standing requirements hold. Furthermore,

$$\epsilon_n^2 = \frac{s_n \log J_n}{n} \asymp N_n^{-2\alpha/(2\alpha+1)}.$$

Choose fixed constants $C_r, C_s, c_e > 0$ such that $\epsilon_n \leq C_r N_n^{-\alpha/(2\alpha+1)}$, $s_n \leq C_s a_n$, and $n\tilde{\epsilon}_n^2 \geq c_e a_n \log J_n$ eventually. If M_0 and A are the constants in Theorem 1, set $M = M_0 C_r$, $A' = AC_s$, and $c = c_e$. The theorem then gives exactly the two displayed expectation bounds and their P_0 -probability consequences. $\square$

Proof of Corollary 2. The restriction on b gives $m_n \leq L_n$ and $a_n \leq J_n$ eventually. Apply (20) with $m = m_n$. The resulting θ_0 obeys

$$\begin{aligned} \|f_0 - f_{\theta_0}\|_{2,n} &\lesssim m_n^{-\alpha} \asymp N_n^{-\alpha/(2\alpha+d)} \leq \tilde{\epsilon}_n, \\ \#\text{supp}(\theta_0) &\leq a_n, \end{aligned} \quad \|\theta_0\|_\infty \leq E_n,$$

after increasing C_ϵ if necessary. Thus Assumption 2 holds.

Since $\log J_n \asymp \log n$,

$$\begin{aligned} a_n &\asymp N_n^{d/(2\alpha+d)}, \\ n\tilde{\epsilon}_n^2 &\asymp nN_n^{-2\alpha/(2\alpha+d)} = N_n^{d/(2\alpha+d)} \log J_n \asymp a_n \log J_n. \end{aligned}$$

It follows that $\tilde{\epsilon}_n \downarrow 0$, $s_n \asymp a_n$, and $s_n \log J_n = o(n)$. Finally,

$$\epsilon_n \asymp \sqrt{\frac{a_n \log J_n}{n}} \asymp N_n^{-\alpha/(2\alpha+d)}.$$

Choose fixed constants $C_r, C_s, c_e > 0$ such that $\epsilon_n \leq C_r N_n^{-\alpha/(2\alpha+d)}$, $s_n \leq C_s a_n$, and $n\tilde{\epsilon}_n^2 \geq c_e a_n \log J_n$ eventually. If M_0 and A are the constants in Theorem 1, set $M = M_0 C_r$, $A' = A C_s$, and $c = c_e$. Applying the theorem with these absorbed constants gives every asserted conclusion. $\square$

Proof of Corollary 3. Equations (24)–(25) give

$$\alpha_k \beta_k = r \quad (k = 1, \dots, d), \quad \sum_{k=1}^d \beta_k = D, \quad 2r + D = 1. \quad (31)$$

The inequalities $b_k \geq \beta_k$ ensure $m_{n,k} \leq L_{n,k}$ for every k and $a_n \leq J_n$ eventually. Apply (26) with $\mathbf{m} = \mathbf{m}_n$. The resulting vector θ_0 satisfies

$$\begin{aligned} \|f_0 - f_{\theta_0}\|_{2,n} &\lesssim \sum_{k=1}^d N_n^{-\alpha_k \beta_k} = d N_n^{-r} \leq \tilde{\epsilon}_n, \\ \#\text{supp}(\theta_0) &\leq a_n, \quad \|\theta_0\|_\infty \leq E_n, \end{aligned}$$

after choosing C_e sufficiently large. Hence Assumption 2 holds.

Because the b_k are fixed, $\log J_n \asymp \log n$. By (31),

$$\begin{aligned} a_n &\asymp \prod_{k=1}^d N_n^{\beta_k} = N_n^D, \\ n\tilde{\epsilon}_n^2 &\asymp n N_n^{-2r} = N_n^{1-2r} \log J_n = N_n^D \log J_n \asymp a_n \log J_n. \end{aligned}$$

Thus $\tilde{\epsilon}_n \downarrow 0$, $s_n \asymp a_n$, $n\tilde{\epsilon}_n^2 \rightarrow \infty$, and $s_n \log J_n = o(n)$ because $D < 1$. Moreover,

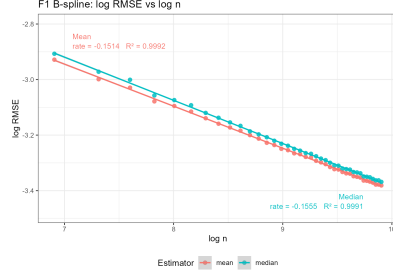
$$\epsilon_n^2 \asymp \frac{N_n^D \log J_n}{n} = N_n^{D-1} = N_n^{-2r}.$$

Choose fixed constants $C_r, C_s, c_e > 0$ such that $\epsilon_n \leq C_r N_n^{-r}$, $s_n \leq C_s a_n$, and $n\tilde{\epsilon}_n^2 \geq c_e a_n \log J_n$ eventually. If M_0 and A are the constants in Theorem 1, set $M = M_0 C_r$, $A' = A C_s$, and $c = c_e$. Applying the theorem with these absorbed constants gives every asserted expectation, posterior-contraction, and effective-support conclusion. $\square$

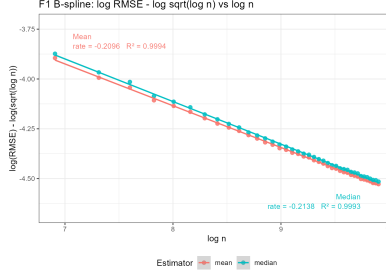
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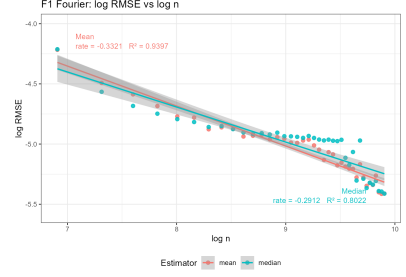
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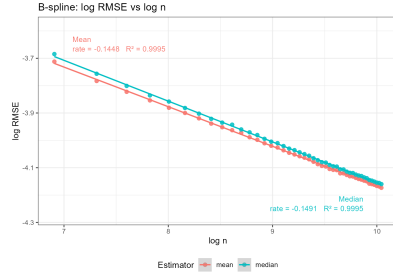
(a) f_1 , B-spline, raw



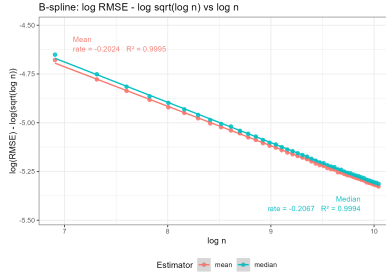
(b) f_1 , B-spline, adjusted



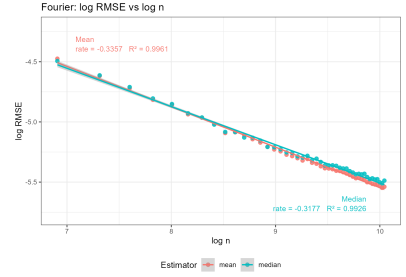
(c) f_1 , cosine, raw



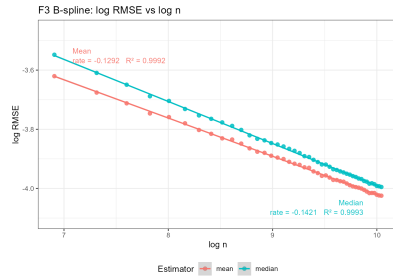
(d) f_2 , B-spline, raw



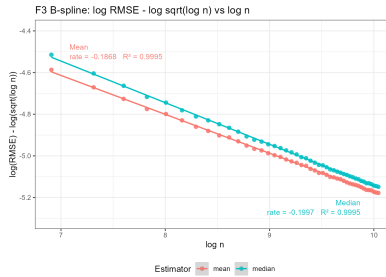
(e) f_2 , B-spline, adjusted



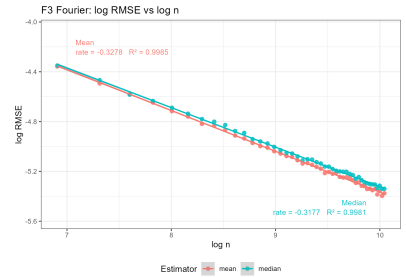
(f) f_2 , cosine, raw



(g) f_3 , B-spline, raw



(h) f_3 , B-spline, adjusted



(i) f_3 , cosine, raw

Figure 2: Empirical RMSE trajectories. The first two columns show the raw and $\sqrt{\log n}$ -adjusted B-spline responses; the third shows the raw cosine response. Salmon and teal correspond to coefficientwise posterior-mean and posterior-median reconstructions, respectively. Each plotted point summarizes the available replications at that sample size by their median.

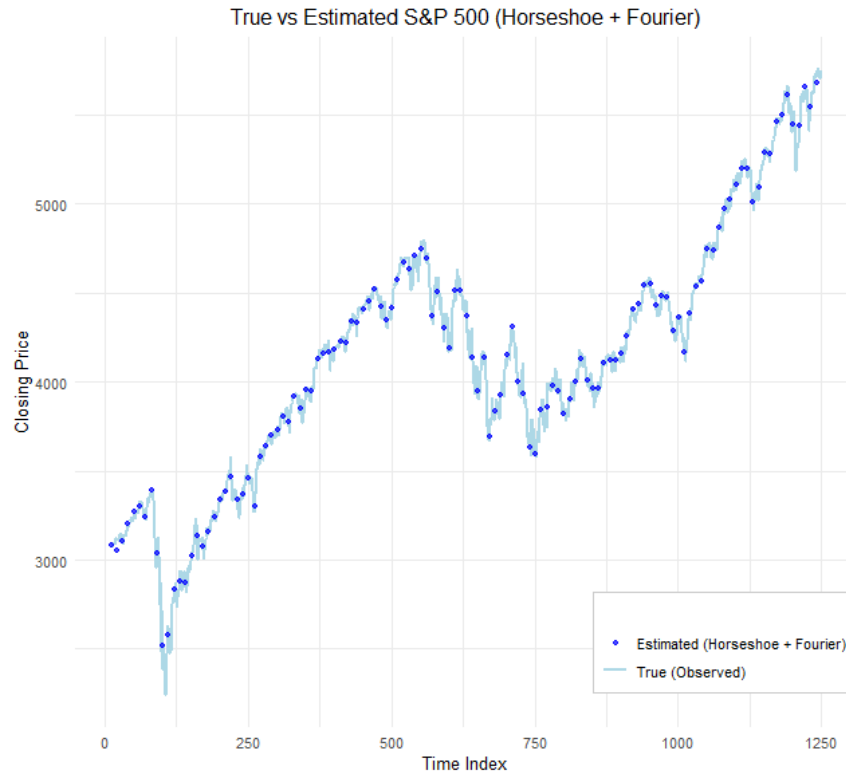


Figure 3: Descriptive Fourier–Horseshoe smooth of the observed S&P 500 closing-price series. The time index is scaled to $[0, 1]$ for fitting and displayed on its original index.

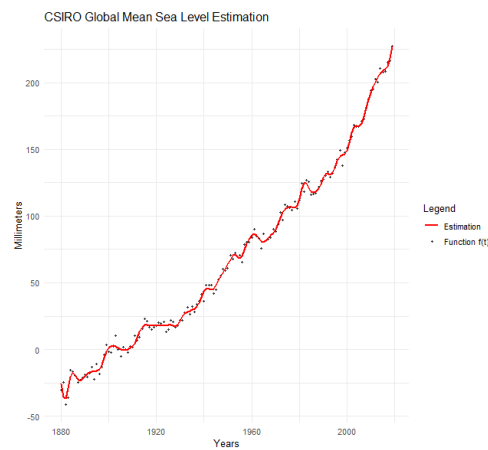


Figure 4: Descriptive cubic-B-spline Horseshoe smooth of the observed GMSL anomalies from 1880 to 2019.